BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022 • Paper Code: DIK-HSSC-XI-2022 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 18 | **Paper Code:** DIK-HSSC-XI-2022 (A)

Q1. Choose the correct option i.e. A,B,C,D and fill it up in the correct bubble in the given MCQs sheet.

Qi. The total number of atoms in 22.4 dm^3 of Co_2 at STP:

(A) 6.02×10^{23}
(B) $2 \times 6.02 \times 10^{23}$
(C) $3 \times 6.02 \times 10^{23}$
(D) $4 \times 6.02 \times 10^{23}$

Qii. Zeeman's effect is explain by:

(A) Principal quantum number
(B) Azimuthal quantum number
(C) Magnetic quantum number
(D) Spin quantum number

Qiii. Which set of quantum number is not correct?

(A) n = 4 l = 2 me = 2 s = +1/2
(B) n = 4 l = 2 me = 2 s = [illegible in source scan]
(C) n = 4 l = 1 me = 0 s = +1/2
(D) n = 4 l = 0 me = 0 s = [illegible in source scan]

Qiv. The molecular orbital with highest energy among the following:

(A) π2Py
(B) π*2Py
(C) δ*2Py
(D) δ2Py

Qv. The volume of given mass of a gas become double when temperature is changed from $50C^0$ to_____ at constant pressure.

(A) $100C^0$
(B) $371C^0$
(C) $373C^0$
(D) 100K

Qvi. The properties of liquid crystal similar to crystalline solids is_____

(A) Optical properties
(B) Some degree of order
(C) Surface tension
(D) Both A and B

Qvii. In body central cubic lattices each lattice point inside the unit cell is shared by_____ unit cell.

(A) One
(B) Two
(C) Three
(D) Eight

Qviii. Weakest force of attraction is found in_____

(A) Covalent solid
(B) Metallic solid
(C) Ionic solid
(D) Molecular solid

Qix. The reaction in which KP < Kc:

(A) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
(B) $PCl_5(g) \rightleftharpoons PCl_3(g) + PCl_2(g)$
(C) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
(D) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

Qx. Weakest London force are found in_____

(A) Kr
(B) Ar
(C) Ne
(D) He

Qxi. The conjugate base of HCO_3^- is _____

(A) CO
(B) CO_3^{-2}
(C) $H_2CO_3^-$
(D) CO_2

Qxii. The solution having highest PH is _____

(A) 0.01MHCl
(B) $0.0001MHNO_3$
(C) 0.01MNaOH
(D) 0.001MNaOH

Qxiii. The reaction given below is an example of: $3O_2(g) \text{—}NO(g)\rightarrow 2O_3(g)$

(A) Homogeneous catalysis
(B) Heterogeneous catalysis
(C) Enzyme catalysis
(D) Uncatalysed reaction

Qxiv. The solution having highest boiling point is _____

(A) 0.01 M glucose
(B) 0.05M sucrose
(C) 0.1 M urea
(D) All have same b.p

Qxv. Heat of solution is the amount of heat released or absorbed when one mole of solute is dissolved in _____ solvent

(A) 100g
(B) 100ml
(C) $1dm^3$
(D) excess

Qxvi. The process for which enthalpy change may be positive or negative?

(A) Combustion
(B) Formation
(C) Neutraliatioin
(D) Ionization potential

Qxvii. Oxidation states of S in SO_2, H_2 H_2SO_4 are___respectively.

(A) +4,+2,+6
(B) +4,−2,+6
(C) −4,+2,−6
(D) +4,−2,−6

Qxviii. Galvanizing is deposition of_____on other metals.

(A) Ni
(B) Cr
(C) Zn
(D) Sn

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 67

Q2. (Marks 40) Answer any TEN questions of the following. All questions carry equal marks: [40]

i. Calculate the volume of NH_3 produced at STP if 10g of H_2 is reacted with excess of nitrogen.

ii. Write any four characteristics of canal rays.

iii. Explain with one example the ionic character of covalent bond.

iv. Rate of effusion of hydrogen gas was found to be 5times to that of an unknown gas at $20C^0$ calculate the molecular mass of the unknown gas.

v. Define and explain viscosity of a liquid. Also write the names of factors which affect viscosity of liquid.

vi. What are the metallic solids? Explain with two examples. Write two properties of metallic solids.

vii. Kp for the following equilibrium established in $1dm^3$ vessel was 0.82 atm at 523k. Calculate Kc and Kx. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

viii. Define and explain levelling effect.

ix. Why Pi(π) bond is weaker than sigma (δ) bond?

x. Define order of reaction. Also define first order, second order and third order of reaction along with their proper representative rate equation.

xi. A solution of 0.1g sucrose was dissolved in 500g of a solvent. Elevation of b.p was found to be $0.2C^0$. Calculate Kb for solvent.(atomic mass=C=12amu, H=1amu, 0=16 amu)

xii. Addition of $50cm^3$ of $2mol/dm^{-3}$ HCl solution caused a rise in temperature of $13.6C^0$ calculate heat of naturalization per mole. Specific heat of water is $4.2\ Jg^{-1}K^{-1}$

xiii. Balance the following redox chemical equation using oxidation number method. $Mg + HCl \rightarrow MgCl_2 + H_2$

SECTION – C

Maximum Marks: 27

Note: Note: Attempt any Three Questions. All questions carry equal marks:

Q.3 a. Define limiting reagent and excess reagent. How limiting reagent is determined in a chemical reaction.

b. Define osmotic pressure. How it is measured experimentally.

Q.4 a. How concept of atomic number was derived from experiments on X-rays.

b. Define a fuel cell. Write construction and working of H_2O_2 fuel cell and balanced chemical equation for redox process.

Q.5 a. Drive the ideal gas equation and relation for the calculation of mass and concentration of ideal gas.

b. Explain crystal lattice of sodium chloride.

Q.6 a. Three mole each of H_2 and I2 were reacted at $460C^0$ to establish equilibrium in $6dm^3$ vessel. Calculate equilibrium concentration of H_2, I2 and HI. Kc = 49 $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$.

b. Define state and state function of a system.